

Quiz1 - Results



Attempt 1 of 1

Written Jan 20, 2026 5:00 PM - Jan 20, 2026 5:08 PM

Attempt Score **10 / 10 - 100 %**

Overall Grade (Highest Attempt) **10 / 10 - 100 %**

Question 1

1 / 1 point

Which of the following tasks is **NOT** a classification task?

- ☒ Predicting house prices based on size, location, and features
- ☐ Determining whether a movie review is positive or negative
- ☐ Predicting the type of object in an image, e.g., cat, bicycle, etc
- ☐ Identifying whether a social media post contains hate speech

Question 2

1 / 1 point

Which of the following statements about bigram features are **CORRECT**?

- ☒ Bigram features use more parameters than unigrams
- ☒ Bigram models can easily overfit the data
- ☐ Bigram features can model sentence structure



Bigram features can represent semantic meaning



Bigram features can outperform unigrams on some tasks

Question 3

1 / 1 point

In Naive Bayes with Laplace smoothing, the parameter λ must be an integer.

☐ True

✓ ☒ False

Question 4

1 / 1 point

In Naive Bayes, why can numerical underflow occur for long documents?



Too many features are removed



Too many small probabilities are multiplied



Too many zero counts



Too many classes

Question 5

1 / 1 point

Given the ground-truth labels [0, 1, 1, 0] and the model predictions [1, 1, 1, 0], what is the precision of the prediction?



1/2



1/4



2/3



3/4

Question 6

1 / 1 point

Why do we need to evaluate models on unseen examples?

- ☐ To minimize the loss function
- ✓ ☒ To measure generalization
- ☐ To increase training accuracy
- ☐ To reduce computation time

Question 7

1 / 1 point

In Logistic Regression, what role do weights play in making predictions?

- ☐ Represent the probability of each class directly
- ☐ Decide how many training examples are used by the model
- ☐ Remove irrelevant features from the dataset automatically
- ✓ ☒ Determine the importance of each feature in predicting the class label

Question 8

1 / 1 point

The sigmoid function maps values from $(-\infty, \infty)$ to approximately $(-1, 1)$

- ☐ True
- ✓ ☒ False

Question 9

1 / 1 point

Naive Bayes assumes word occurrences are independent when conditioned on the class.

- ✓ ☒ True
☐ False

Question 10

1 / 1 point

Which of the following statements about preprocessing is **WRONG**?

- ☐ Tokenization can split words and punctuation
- ✓ ☒ Lowercasing text does not remove any meaning
- ☐ Removing external URL links often removes unhelpful text
- ☐ Stemming and lemmatization reduce vocabulary size

Done